



Database Systems:

Individual Assignment

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R1:

a. Detailed relational Schema

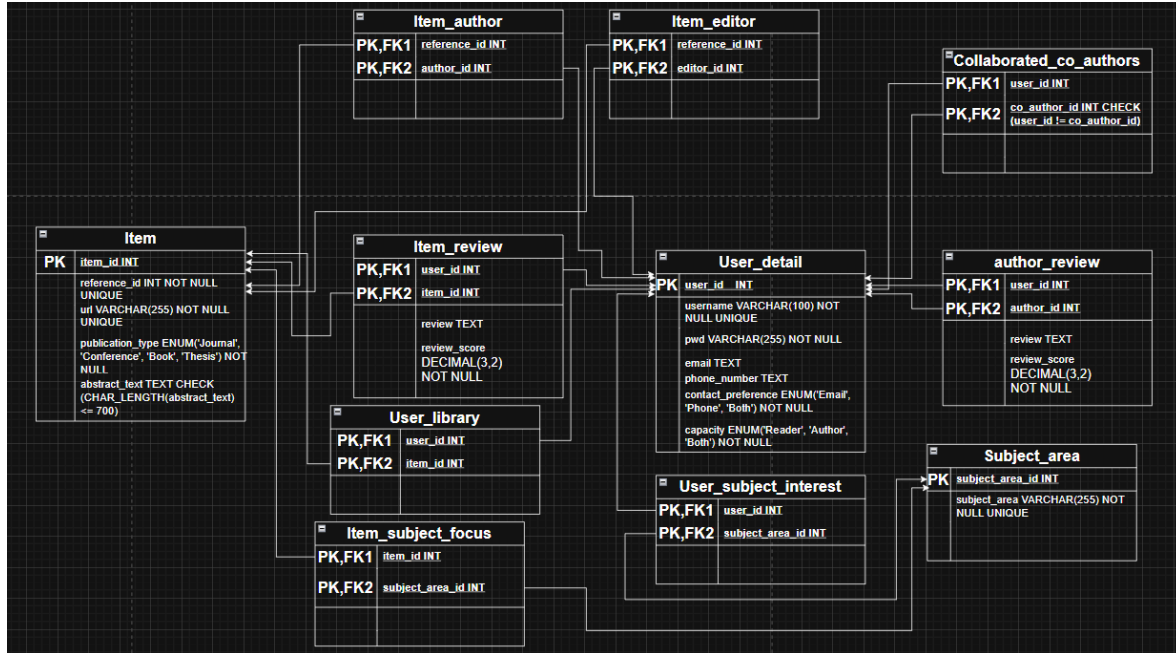


Figure 1: Relation schema – User and Item entity

This relational schema encompasses the **User** entity, along with its associated attributes and its relationship with the **Item** entity.

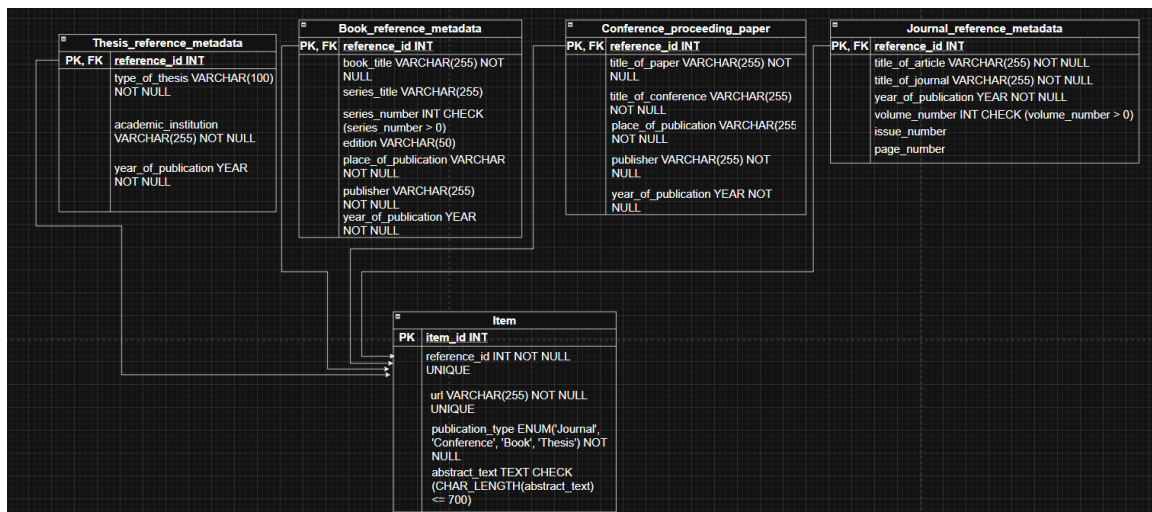


Figure 2: Relation schema – Item and meta data entity

This relational schema represents how the **Item** entity associates its metadata with respect to the type of publication (Journal, Conference, Book, Thesis).

b. Design decision and references

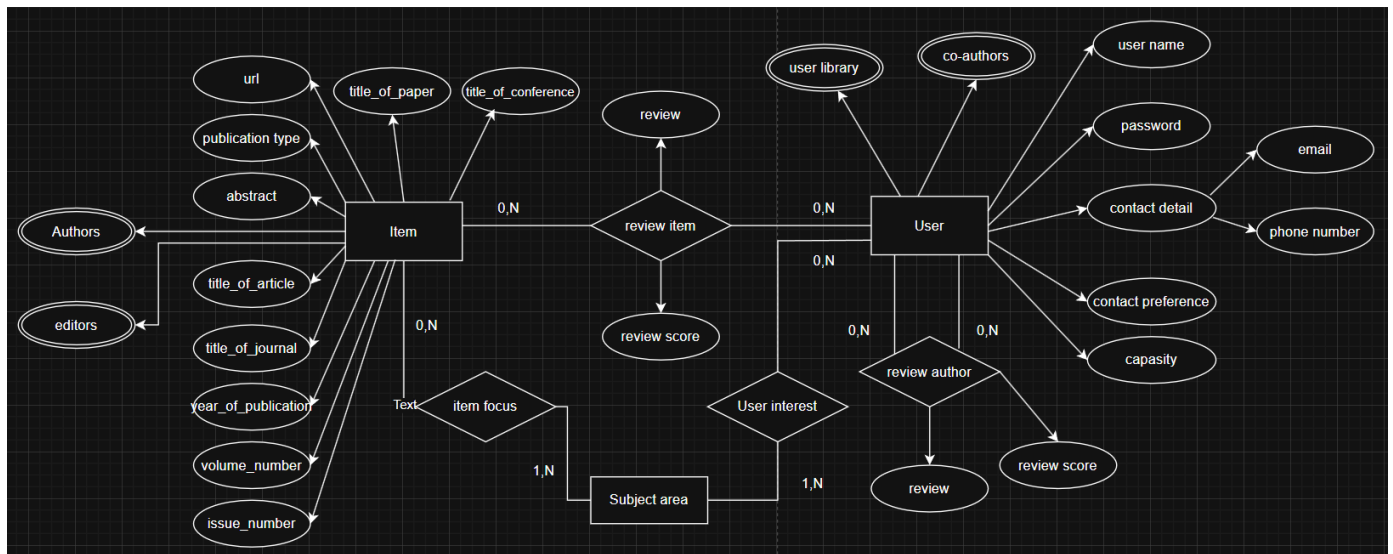


Figure 3: ER diagram

Reference 1: “We recently secured funding to implement a new that will recommend appropriate articles and books to our users every time they login into our system. For instance, every recommended item would have to represent a good match for a user based on their scientific interests”

Conclusion 1: The database design should include three entities User (registered user of the system), Item (research works) and Subject interest.

Reference 2: “Primarily, every item, e.g., a journal article, needs to be stored in the database you will be creating. Note that we do not keep the actual content of every item, i.e., text, tables, figures, references etc. Instead, the database should store only metadata of every item, some statistics, reference information and hyperlinks to the primary sources” and “Information for a published piece of work consists of its title, the corresponding author, a list of co-authors if they exist, the type of publication, the scientific subject or subjects the work focuses on and the abstract text (max 700 words)”

Conclusion 2: The Item table includes reference information, abstract text (max 700 words) and hyperlink as attributes.

Reference 3: “Oh, and let's not forget reviews and reviewing scores for items and authors.” and “whether a user is a reader, an author or both. In both cases, though, the user should be able to review items and score them accordingly.” and “a user may submit only one review and a single score for a particular item.” and “Moreover, we want users to review what they read, such that every item has a score based on readers' appreciation (average score) and a list of reviews”

Conclusion 3: The Item user table has 2 relationships. A recursive relationship with itself via review-author relationship where user can review author (Author also belongs to user) and review-item relationship with item table. Cardinality of review-author is (0, N) -(0, N) and review-item is (0, N) - (0, N). review and review score are descriptive attributes of both relationships.

Reference 4: “Oh, yes! An item can either be a journal article, conference paper, a book or a university thesis. The database should be able to reflect those different types and the specifics in producing references. We will provide you with a table (see Table A) with the minimum details the database should be able to handle.”

Conclusion 4: The item table has an attribute publication type only belongs to these 4 categories (Journal article, Book, conference paper, Thesis). With respect to the type of the publication, reference meta data varies according to the annex table.

Reference 5: Table A

Conclusion 5: Item has multi valued attributes include Author and editor with respect to the publication type.

Reference 6: “True. An article may make a new contribution to the area of mobile robotics by addressing group behaviour using evolutionary computation mechanisms. In that case, the article relates to both "Mobile Robotics" and "Evolutionary Computation". The important thing here is that users define their interests in relation to available subjects in the system and therefore, the latter should make appropriate recommendations.”

Conclusion 6: The identified entity Subject interest formulated a table and kept 2 relationships with Item and User tables. A relationship with Item table via Item_subject_focus relationship and User_subject_interest relationship with User table. Cardinality of Item_subject_focus is (0, N) -(1, N) and User_subject_interest is (0, N) -(1, N).

Reference 7: “We only need generic data, such as the account's user name and password, contact details and contact preference, as well as their capacity in the system, i.e., whether a user is a reader, an author or both. Most importantly, the database should keep track of all items a user has read. Conceptually, this can be seen as the user's library.

Conclusion 7: User table single attributes are username, password, contact details (composite attributes) preference and capacity. User table has another multivalued attribute user library.

Reference 8: “but other things should count too. If a user happens to be an author that has collaborated with others in the writing of an article, recommendations for her co-authors' items should be prioritised.”

Conclusion 8: If a user is an author, then they also associate with another multi valued attribute co-author.

ER to logical design formulation with cardinality consideration

 Blue: Denotes foreign keys

 Red: Highlights attributes with constraints

Underlined: Represents primary keys in the schema.

User (user id, username, pwd, contact detail, contact preference, **capacity (includes 3 category)**)

Collaborated_co_authors (user id, co-author id)

User_library (user id, item id, **item url**)

Subject_area (subject area id, **subject area**)

User_subject_interest (user id, subject area id)

Item_subject_focus (item id, subject area id)

Item_review (user id, item id, **review**, **review score**)

Author_review (user id, author id, **review**, **review score**)

Item (item id, url, **publication type (only belongs to 4 categories)**, **abstract**, **reference id**)

Journal_reference_metadata (reference id, title of article, title of journal, year of publication, **volume number (includes > 0)**, **issue number**, **page number**)

Item_author (reference id, author id)

Conference proceeding paper (reference id, title of paper, title of conference, place of publication, **publisher**, **year of publication**)

Item_editor (reference id, editor id)

Book_reference_metadata (reference id, **book title**, **series title**, **series number**, **edition**, **place of publication**, **publisher**, **year of publication**)

Thesis_reference_metadata (reference id, **type of thesis**, **academic institution**, **year of publication**)

c. Create database schema and populate records

Attached zip file (Schemas) includes separate SQL script for each table. SQL script contains 3 parts

- I. Table creation - DDL (Data Definition Language) command
- II. Record insertion – DML (Data Manipulation Language) command

d. constraint validation

- III. Constraint validation - DML (Data Manipulation Language) command

R2:

e. list of all identified business processes grouped by the user type

user group1 Readers:

1. Calculate match score – base score calculation irrespective to the subject interest
2. Calculate match score – if the item is associated with subjects that match the interests of the user
3. Calculate match score – based on the user library

user group2 Authors:

1. Calculate match score – base score calculation irrespective to the subject interest
2. Calculate match score – if the item is associated with subjects that match the interests of the user
3. Calculate match score – based on the user's co-authors that appear in the item's author list
4. Calculate match score – based on the user library

f. Recommendation logic SQL query

Attached zip file (Recommendation-logics) includes separate SQL script for each user group (Reader and Author) based on the identified business processes.

R3:

g. Kimball's dimensional modelling

Recommendation modelling

Datamart 1: Calculate match score – base score calculation irrespective to the subject interest (Business process 1)

Grain: Base score per user per item

Dimensions: dim_user, dim_item, dim_Item_review

Facts: base score

Fact table: fact_user_item_base_score

Datamart 2: Calculate match score – if the item is associated with subjects that match the interests of the user (Business process 2)

Grain: Subject match score per user per item

Dimensions: dim_user, dim_item, dim_user_subject_interest,
dim_item_subject_focus

Facts: subject match score

Fact table: fact_user_item_subject_match_score

Datamart 3: Calculate match score – based on the user library (Business process 3)

Grain: Library match score per user per item

Dimensions: dim_user, dim_User_library, dim_item, dim_item_author,
dim_item_subject_focus

Facts: library author match score, library subject match score

Fact table: fact_user_library_match_score

Datamart 4: Calculate match score – based on the user's co-authors that appear in the item's author list ((Business process 4)

Grain: Co-author match score per user per item

Dimensions: dim_item, dim_user, dim_item_author, dim_collaborated_co_authors

Facts: Co-author match score

Fact table: fact_user_co_author_match_score

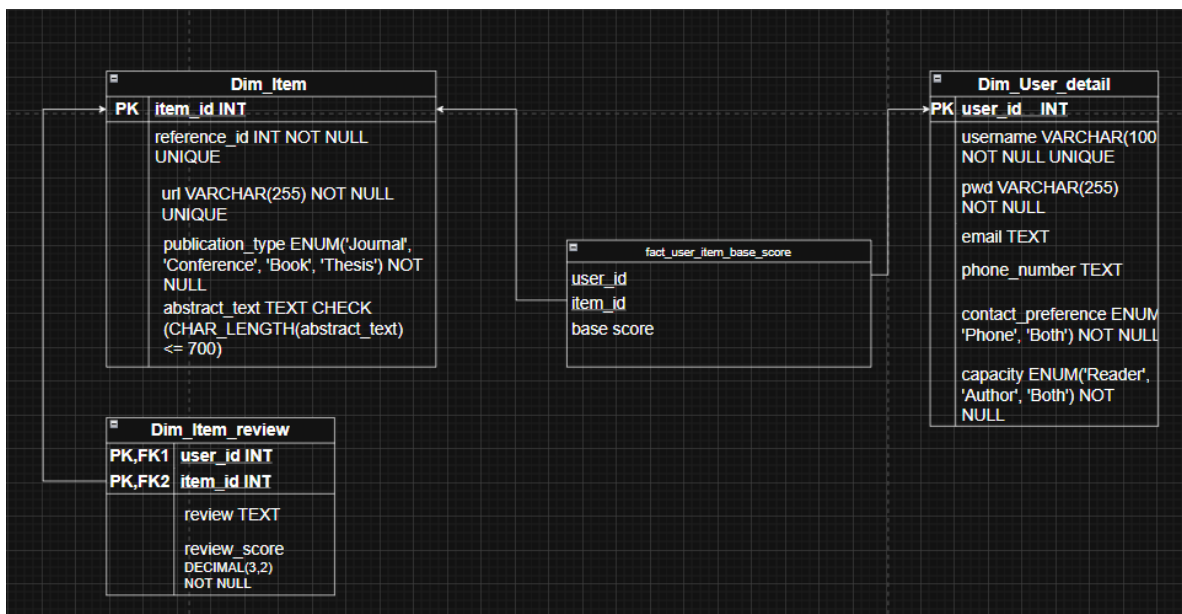


Figure 4: Star schema Datamart 1

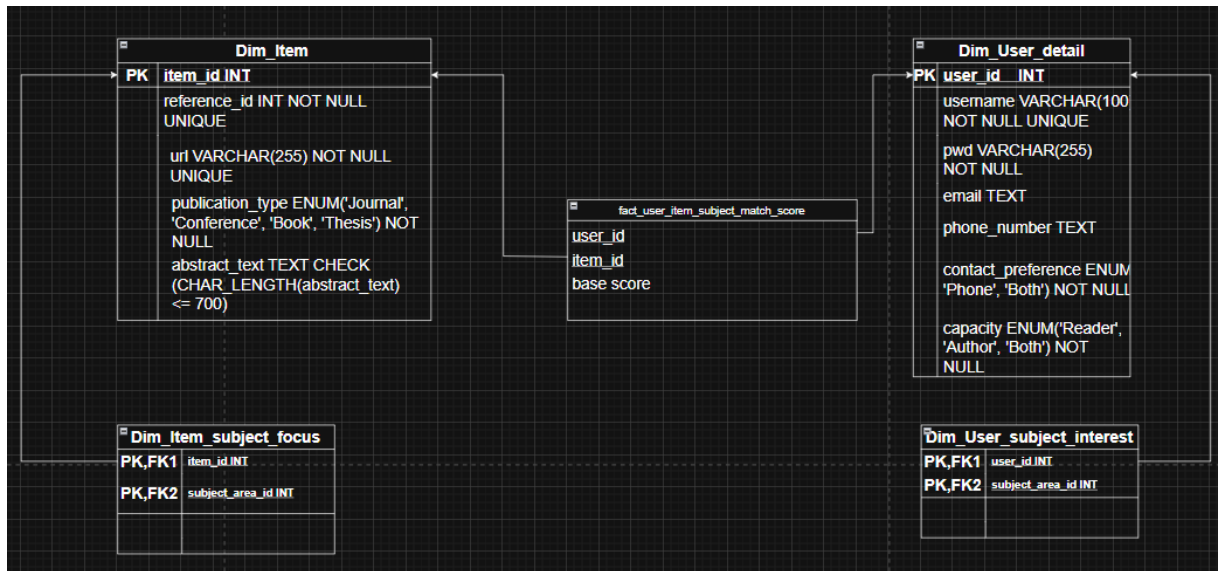


Figure 5: Star schema Datamart 2

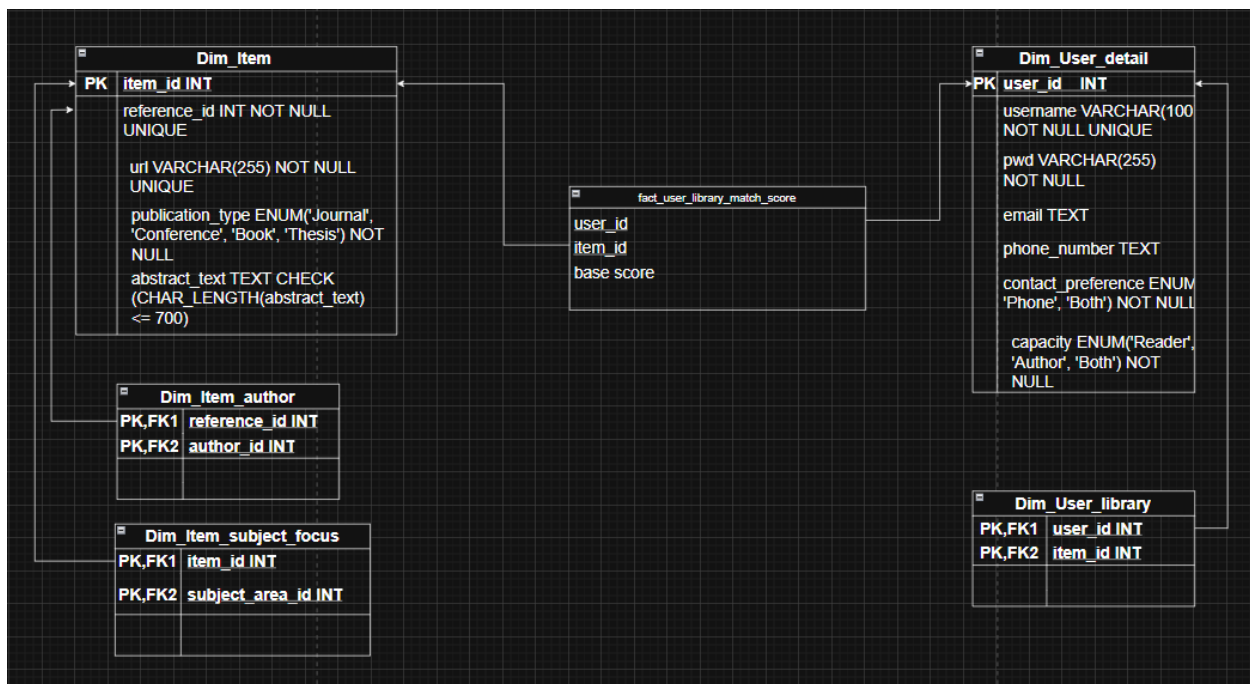


Figure 6: Star schema Datamart 3

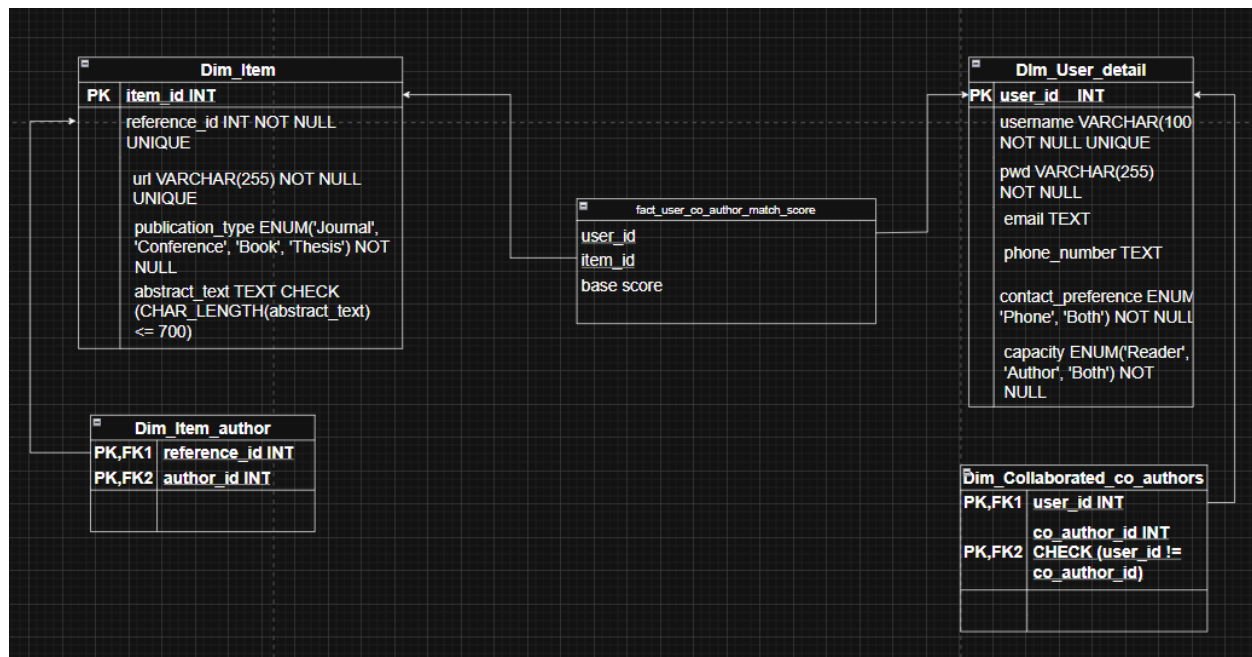


Figure 7: Star schema Datamart 4

References

Sawyer, A. (2024, July 29). *Kimball Star Schemas in Data Warehousing: Part 3 – Mastering Bridge Tables for Many-to-many Relationships*. Data Training Lab.
<https://blog.datatraininglab.com/kimball-star-schemas-in-data-warehousing-part-3-048a12b061c9>